



EUROPE

China's strategic petroleum reserve as a geoeconomic tool

A 'what if' exercise

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Published by the RAND Corporation, Santa Monica, Calif., and Cambridge, UK

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1. About this report

China's Strategic Petroleum Reserve (SPR) was established in March 2004, with the aim of constructing China's first oil base to protect the country's energy security.¹ Since then, China's SPR has evolved from a small coastal network into a nationwide system of large-scale storage facilities combining central government, commercial and provincial reserves. As of May 2026, the Chinese SPR is estimated to hold between 1.1 and 1.4 billion barrels of oil, equivalent to around 110–140 days of net imports.² The US SPR, in comparison, currently comprises 413 million barrels, equivalent to approximately 125 days of net imports.³ A key distinction between the two nations is the treatment of government- versus corporate-held petroleum inventory. In China, the corporate-held inventory carries a mandate of state responsibility that allows government intervention in a national emergency, effectively giving the government control over both commercial and government reserves.

While the Chinese SPR is primarily designed to mitigate risks from oil supply chain disruptions, its size compared to that of other countries could give China a large potential leverage on international energy markets and, by extension, in the global economy. In other words, China's SPR could evolve from an economic policy device to a geoeconomic tool allowing the country to achieve goals such as **exercising leverage on other nations, protecting national security, influencing alliances, punishing adversaries and affecting global supply chains**.

The value of this leverage is likely to increase if reserve releases by International Energy Agency (IEA) member states, such as the March 2026 release of 400 million barrels of oil, fail to normalise volatility in the oil markets,⁴ which is a possible outcome according to experts.⁵ Unlike the United States, China is not a member of the IEA and before the 2026 Iran War, the combined SPR of the IEA member states and China's SPR were of comparable size.

Nevertheless, Chinese policy around the use of their SPR as a geoeconomic tool, particularly in crisis scenarios, is not well understood. Although China has built its SPR as a resilience tool, **the size of the SPR compared to other global oil reserves could give China the leverage to use it in a more strategic way**.

¹ Ministry of Ecology and Environment (PRC) (2007).

² Meidan (2026); see also US Energy Information Administration (2026a).

³ International Energy Agency (IEA) (2026a).

⁴ International Energy Agency (IEA) (2026b). Volatility refers to sudden swings in oil prices in international markets. For instance, since the beginning of the US-Iran war, the price of West Texas Intermediate (WTI) oil has fluctuated widely above and below \$100/barrel.

⁵ O'Brient (2026).

This project investigates potential ways in which China could do so. Our analysis, however, does not explore whether China actually intends to use SPR as a geoeconomic tool. Specifically, we are interested in evaluating the four factors below:

- What does China's SPR look like and how is it managed?
- Could China use its SPR to achieve geoeconomic effect?⁶
- Under what scenarios might China use its SPR as a geoeconomic tool?
- How should allied nations or partners respond to specific geoeconomic applications of China's SPR?

China Research Center

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Funding

This research project was independently initiated and conducted within the China Research Center using a gift from philanthropist Michael Tang, as well as gifts from other RAND supporters and income from operations. RAND donors and grantors have no influence over research findings or recommendations.

Acknowledgements

We thank the RAND China Research Center, and in particular Jude Blanchette for his support of this project, as well as our QA reviewers Fiona Quimbre and Francesca Ghiretti for their insightful comments.

⁶ Geoeconomic effect involves using economic power to project influence and secure strategic interests. An example is the US economic sanctions on Russia in 2022, which aimed to influence Russian strategic behaviour.

2. Methodology

To answer the research questions set out above, we followed a mixed methods approach:

- **Technical and policy assessment of China's SPR.** We used a short and targeted literature review to generate a rapid evidence assessment to characterise the technical attributes of China's SPR (e.g. volume, stockpiling strategy, etc.) as well as the policy mechanisms that regulate its use.
- **Rubric for evaluating China's SPR potential for use as a geoeconomic tool.** Based on a rapid literature review, we identified the necessary attributes of a geoeconomic tool. These attributes were then represented in a rubric which we used to assess the Chinese SPR's geoeconomic potential.
- **Vignette development.** Using a Political Economic Social Technical Legal and Environmental (PESTLE) classification,⁷ we identified six enabling drivers (which may encourage the use of the SPR to further geoeconomic goals) and six restricting drivers (which may limit the use of the SPR as a geoeconomic tool). We then drafted a longlist of eight vignettes outlining different ways in which China could leverage its SPR, which we presented to a group of four RAND subject-matter experts with a background in macroeconomics, China studies and international relations. In the vignettes, we considered a time horizon of five years, which aligns with the principal cadence of Chinese strategic planning. In practice, major priorities in energy security, industrial policy and foreign economic strategy are articulated and operationalised through China's Five-Year Plans. Based on the input of the subject-matter experts, we refined the longlist to three plausible vignettes outlining how China may leverage its SPR.

⁷ Makos (2024).

3. Understanding China's SPR

3.1. Technical overview

Even though China's SPR is relatively young compared to other oil reserves, it has rapidly expanded in size and strategic reach. The overall concept of SPRs was originally established as a coordinated means to ensure national security and economic stability during geopolitically induced supply shocks. Following the Arab oil embargo (1973–4), the IEA mandated a 60-day (later updated to 90-day) net-import equivalent stockpile for its member countries.⁸ Two decades later, after becoming a net importer of crude oil in 1993, China started planning for its own 90-day SPR, without joining the IEA, as part of the long-term objectives of its Ninth Five-year Plan in 1996.⁹

Today, China has the world's largest SPR, estimated to have reached up to 1.4 billion barrels, or 110–140 days net-import equivalent, in December 2025.¹⁰ Figure 1 shows a comparison of the Chinese SPR with that of other countries. One important feature of the Chinese SPR is the distinction in treatment of government reserves (orange) and enterprise reserves (green). The latter are held by commercial firms, but in China they can still be mobilised under national rules stipulated in the 2025 Energy Law (see Section 3.2).¹¹ This arrangement offers Beijing the flexibility to use enterprise reserves as a first line of response, preserving government-owned strategic stocks for larger disruptions; it is also why the IEA counts both in the total Chinese SPR, compared to the 413 million barrels held by the US government SPR. This layered architecture may increase China's capacity to intervene without immediately signalling a full-scale emergency drawdown.¹² Notably, both the US and China are net importers of raw crude oil, with China having surpassed the US as the largest absolute importer in 2017¹³; India, South Korea, Japan and individual

⁸ US Department of Energy (2026).

⁹ Li, H., et al. (2017).

¹⁰ US Energy Information Administration (2026b).

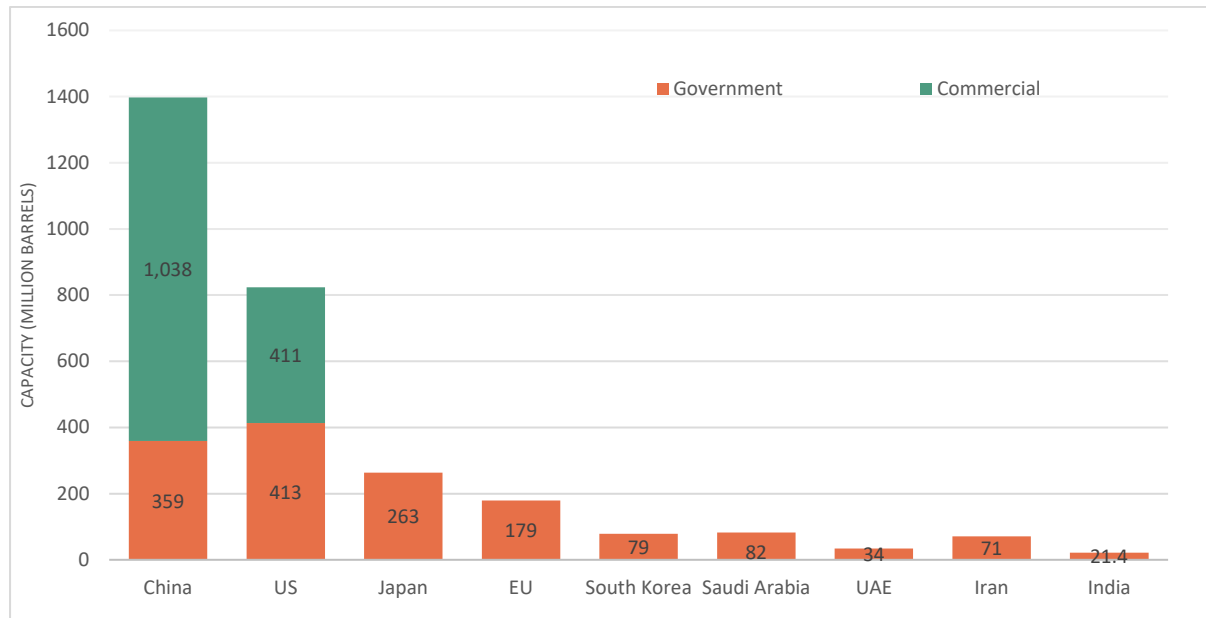
¹¹ The authority to mobilise enterprise-held reserves is provided for in the 2025 Energy Law (Article 54), which empowers the state to direct the use of such reserves in response to supply shocks or major market fluctuations. Supporting provisions in the Energy Law (Article 55) also leave room for interpretation, when the scope of the mobilisation and compliance responsibility are described to cover not only users and enterprises in the energy section (能源企业、能源用户), but also other relevant individuals and entities (其他有关单位和个人).

¹² Bloomberg News (2026).

¹³ Even though the US is currently the world's largest oil producer, it remains a net importer of crude oil. US Energy Information Administration (2018).

countries in the European Union (EU) are the next largest importers.¹⁴ Meanwhile, Saudi Arabia, the United Arab Emirates and Iran are net exporters of raw crude oil.¹⁵

Figure 1. Crude oil inventories (December 2025)



Source: US Energy Information Administration (2026a).

China has used periods of low oil prices and its position as a major buyer of discounted, often sanctioned, crude oil to fill its SPR. Crude imports increased by 4.5 per cent from 11.1 million barrels per day in 2024 to 11.6 million in 2025.¹⁶ Figure 2 shows the top importers of crude oil to China. Middle Eastern imports represent a 41 per cent share of imports, mainly comprised of medium-to-heavy, sour crude types. Russian imports represent 18 per cent, comprised of light, sweet crude from the Eastern Siberia-Pacific Ocean (ESPO) pipeline and a heavier blend from the Urals.¹⁷ Other sources include Brazil (8 per cent), Angola (5 per cent), and Canada (3 per cent).¹⁸ Around 22 per cent of reported 2025 imports are estimated to have been sanctioned – with many Iranian and Venezuelan barrels likely re-exported via Malaysia, for example.¹⁹ Domestic crude is often medium-to-heavy with high wax content, motivating China's extensive heavy refining capabilities. Since 2005, pockets of light crude deposits have also been discovered within China, for example in Bohai Bay Basin.²⁰

¹⁴ World Integrated Trade Solution (2024).

¹⁵ Organization of the Petroleum Exporting Countries (OPEC) (2026).

¹⁶ Li, S., & Jackson (2026).

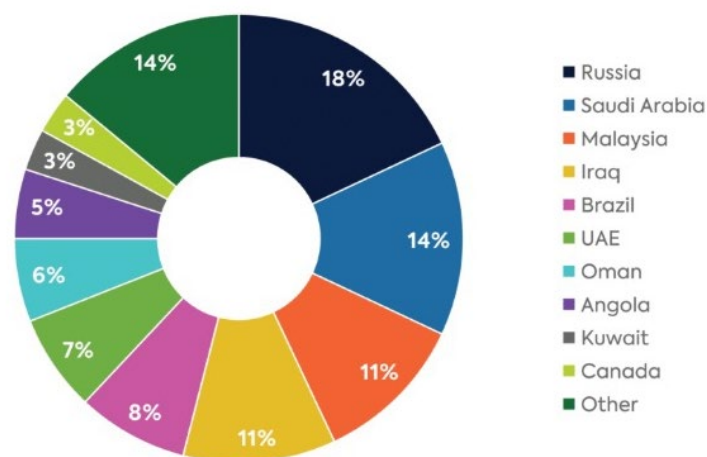
¹⁷ 'Heavy' versus 'light' refers to the density of crude oil, where heavier oils are thicker and tend to yield more residual products, while lighter oils flow more easily and produce more gasoline and diesel. 'Sweet' versus 'sour' describes sulphur content, where sweet crudes have lower sulphur and are easier and cheaper to refine, while sour crudes contain more sulphur and require additional processing; Down (2026).

¹⁸ Down (2026)

¹⁹ Down (2026).

²⁰ Zhao et al. (2013).

Figure 2. Top crude oil importers to China (2025)

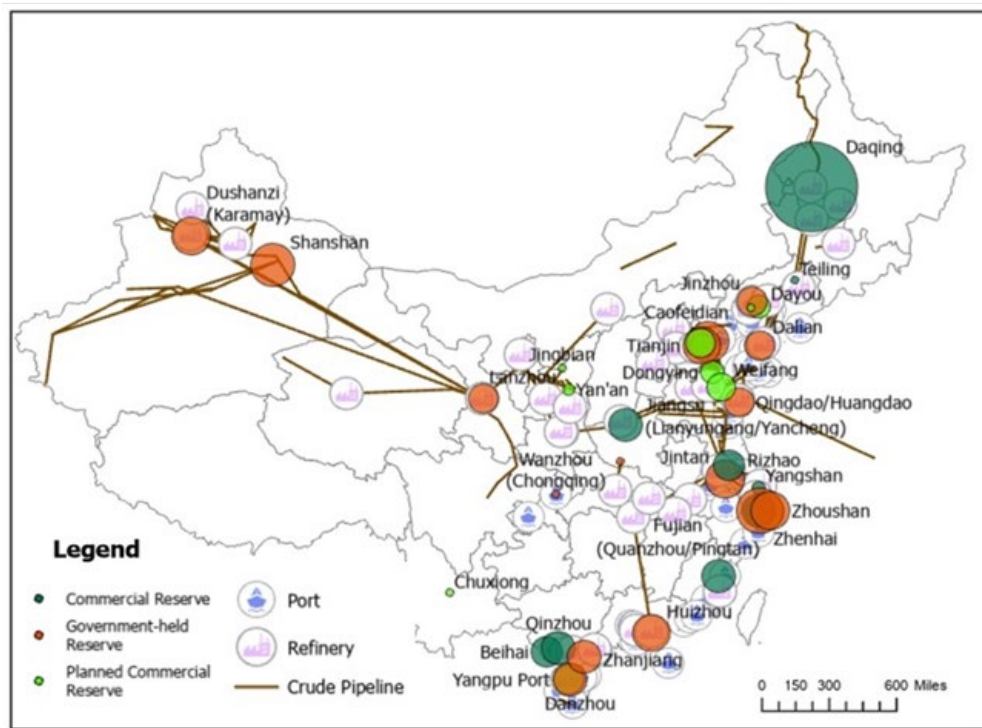


Source: Down (2026).

Collating data from a variety of public sources, Figure 3 illustrates how the SPR has grown physically from four coastal facilities completed in Dalian, Qingdao, Zhoushan and Zhenhai in 2009 to a national network of at least 22 above-ground and underground facilities as of 2023. Coastal facilities hold a mix of commercial (green) and government (orange) facilities close to ports of entry. Inland sites are predominantly government-controlled reserves positioned to stabilise local supply disruptions or commercial reserves located near major oilfields. Eleven commercial sites are planned to be added by 2026 (light green) concentrated along the eastern central and lower coast.²¹

²¹ Reuters (2025).

Figure 3. SPR sites detected in China



Source: Li, H., et al. 2017; Pipeline, ports and refineries from ArcGIS Online.

3.2. Policy overview

China's SPR has evolved from a narrowly defined emergency stockpile into a broader instrument of energy security, macroeconomic management and state capacity. Initially established to reduce vulnerability to external supply disruptions, the SPR now sits within a reserve system combining central government stocks, enterprise-held reserves, and commercial inventories. This transition matters because it gives Beijing more than one pathway to influence domestic and potentially international oil markets: it can build stocks, rotate them, release them selectively, or direct enterprise reserves before drawing on central government holdings.²²

China's SPR differs from the classic IEA model in two respects.²³ Firstly, it is embedded in a more centralised political system in which stockpiling and release decisions can be tied closely to state objectives. Secondly, Chinese policy has framed reserves as both an emergency buffer and a tool for market stabilisation and macroeconomic control. Public messaging around the 2021 crude oil reserve auctions, for example, indicated that the state viewed reserve releases as a way to reduce pressure from rising raw material prices

²² Beijing's ability to manage oil reserves in these ways can be limited by infrastructure capacity, legal restrictions on intervening in enterprise stocks, and the need for coordination between government and private actors.

²³ National Food and Strategic Reserves Administration (PRC) (2018; 2021).

and support domestic refiners.²⁴ This suggests that China manages its SPR as an economic policy tool to shape domestic price conditions and signal state intent to the market.²⁵

A major recent development is the entry into force of the Energy Law of the People’s Republic of China on 1 January 2025, which provides the statutory basis for China’s energy reserve system. The law formalises an integrated structure of government reserves and enterprise reserves, authorises state action in response to supply shocks and major market fluctuations, and strengthens the legal basis for requiring firms to comply with reserve obligations and mobilisation orders. Prior to this, the policy framework relied heavily on draft regulations, administrative guidance and planning documents, especially the 2016 Draft Regulations on National Petroleum Reserves and sectoral guidance issued by the National Development and Reform Commission (NDRC), the National Energy Administration (NEA), and the National Food and Strategic Reserves Administration (NFSRA).

Table 1. Main features of China’s SPR policy framework

Dimension	Main features	Relevance for geoeconomic use
Primary purpose	Energy security and supply disruption management	Provides the core national security justification for maintaining large stocks
Expanded purpose	Market stabilisation and macroeconomic control	Creates policy space for peacetime interventions, including price management
Legal basis	2025 Energy Law; earlier draft regulations and administrative guidance	Strengthens formal authority to mobilise reserves and direct firms
Reserve structure	Government reserves plus enterprise/social responsibility reserves	Enables layered and selective use of stocks
Decision making	Centralised, led by the State Council with NDRC/NEA planning	Facilitates rapid alignment with wider state objectives
Execution	NFSRA management, SOE (notably CNPC, Sinopec and CNOOC) infrastructure and logistics	Gives the state operational reach through state-linked market actors
Operational tools	Stockpiling, rotation, auctions, emergency drawdown	Provides several channels for market influence short of a declared emergency
Transparency	Limited public disclosure of total inventory and release logic	Increases uncertainty and amplifies signalling effects

China appears to rely on stock rotation as a normalised policy tool.²⁶ This mechanism can serve several purposes simultaneously: preserving stock integrity, reducing storage costs, easing domestic supply pressure, and sending a signal to traders and refiners. The 2021 Dalian crude auction is the clearest documented

²⁴ According to the National Food and Strategic Reserves Administration, the release of crude oil reserves was intended ‘to ease the pressure of rising raw material prices for key domestic enterprises’ (‘为重点企业缓解原材料价格上涨压力’), and ‘to ensure the smooth operation of the economy’ (‘保障经济平稳运行’). See National Food and Strategic Reserves Administration (PRC) (2021).

²⁵ National Energy Administration (PRC) (2024).

²⁶ The strategic reserve in China is limited in transparency. Unlike some IEA members, China does not publish official data on total strategic stocks regularly. Estimates of stock levels rely on customs data, tanker tracking, satellite imagery and secondary analysis.

example of the mechanism in action.²⁷ While modest relative to China's estimated stockpile, the release showed China's willingness to use state-held crude to address domestic cost pressures and shape expectations.²⁸

²⁷ According to the National Food and Strategic Reserves Administration, the September 2021 auction released approximately 738,000 tons (about 7.38 million barrels) of crude oil from the Dalian reserve base. The official rationale was 'to ease the pressure of rising raw material prices for key domestic enterprises and ensure the smooth operation of the economy' ('为重点企业缓解原材料价格上涨压力, 保障经济平稳运行'). This was the first time China publicly organised such an auction, reflecting a shift towards using the SPR as a tool for market stabilization in response to international oil price volatility. See National Food and Strategic Reserves Administration (PRC) (2021).

²⁸ Cang & Blas (2021).

4. SPRs as a geoeconomic tool

Geoeconomic tools are economic instruments that governments leverage to advance geopolitical or strategic goals.²⁹ Examples include China's Belt and Road Initiative (BRI), in which the building and financing of infrastructure projects is used by China to achieve strategic objectives.³⁰ China's past use of strategic commodities suggests it is not opposed to using the SPR geoeconomically in a selective and state-directed way. For example, China used its grain reserves in the 1950s–1970s for international diplomacy despite general food shortages.³¹ More recently, rare earth export restrictions during the 2010 dispute with Japan showed how Beijing can use commodity control for diplomatic leverage.³² SPRs have also been used to achieve macroeconomic, geoeconomic and geopolitical objectives (Table 2).

Table 2. Examples of the use of SPRs for macroeconomic, geoeconomic and geopolitical purposes

Event	Geoeconomic impact	Country
2011 Libyan Civil War	SPR released in coordination with IEA, offset supply loss, supported allied energy security	United States & IEA members
2022 Russia-Ukraine War & Energy Crisis	Released record volumes to limit global price surge, shield allies, and pressure Russian revenues	United States & IEA members

Table 3 includes a rubric with attributes usually associated in the literature with a geoeconomic tool that are also present in the case of China's SPR.³³

²⁹ Blackwill & Harris (2016).

³⁰ Rolland (2017).

³¹ Kasahara & Bingjing (2020); Musabelliu (2019); Frank et al. (2025).

³² Baskaran & Dady (2026).

³³ Farrell & Newman (2019).

Table 3. China's SPR as a geoeconomic tool rubric

Attribute	Definition	China's SPR
Economic leverage	The ability to influence other countries or global markets using economic means.	China has the largest SPR in the world.
Strategic flexibility	Capacity to deploy or withhold a resource to advance geopolitical objectives.	China can act quickly during energy crises or geopolitically sensitive moments or decide to preserve its SPR.
Deterrence	The potential to discourage undesirable actions from other actors through the threat or use of economic measures.	The existence of a large SPR lessens China's vulnerability to energy-related coercion, such as embargoes or sudden supply disruptions from major oil-exporting countries.
Resilience enhancement	Strengthening a country's ability to withstand external economic shocks or coercion.	The reserve ensures China has capacity to better endure supply disruptions, mitigating impact on its economy from global energy shocks and thus providing the stability needed for continued economic growth and development.
Policy signalling	Using economic tools to communicate intentions, resolve or capabilities to international actors.	By publicly announcing additions to its SPR or allowing releases during crises, China can signal its preparedness and resolve to both allies and adversaries, demonstrating its capacity to manage energy risks and pursue independent energy policies.

5. Vignette analysis

To examine the different strategic ways in which China could deploy its SPR, we developed a number of vignettes. Vignettes are brief descriptions of hypothetical situations that are presented to knowledgeable individuals, who are then asked to share their normative judgements, views and assessment of them.³⁴ To draft the vignettes, we selected 12 drivers based on the PESTLE framework: six enabling drivers, which could encourage China to use the SPR in a geoeconomic way, as outlined in Table 3 above, and six restricting drivers, which could limit China's use of the SPR as a geoeconomic tool (see Annex).³⁵ Based on these drivers, we created eight vignettes that we presented to a panel of four RAND subject-matter experts with a background in macroeconomics, China studies and international relations, during a two-hour workshop. The experts were asked to comment on the vignettes and their underlying drivers and discuss why a certain vignette would or would not represent a plausible policy decision for the Chinese central government.³⁶ The experts considered three vignettes to represent plausible policy outcomes:



Strategic caution

In the face of mounting global volatility, China adopts a cautious approach towards its SPR. Domestic policymakers calibrate reserve usage to shield the nation from external shocks, such as rising oil prices, sanctions and regional instability. Instead of large-scale releases, authorities allow state refining companies to tap commercial reserves at a modest pace. Some level of higher prices and public discontent are accepted but tightly monitored. While this approach frustrates some industries seeking relief, it reassures the public and global partners that China remains committed to resilience and stability, preserving its energy security amid an increasingly volatile world.

Expert assessment

The subject-matter experts considered this to be the most likely scenario, and most in line with China's focus on energy security and resilience, favouring cautious and limited SPR releases to preserve energy security amid uncertainty.

³⁴ Vignette analysis is a qualitative research method which has been used across a range of social science, education, healthcare and policy analyses. See Hughes et al. (2008); Tremblay et al. (2022); Verneuer-Emre (2022); Almlöf (2023).

³⁵ Enabling drivers: Sanctions and Signalling Resilience; Global Energy Market Influence; Energy Portfolio Diversification; Regional Security Signalling; Strategic Industry Prioritisation; Strategic Partnerships with the Global South. Restricting drivers: Long-term Security of Supply Challenges; Domestic Stability and Public Discontent; Weakening Economic Conditions; Market Volatility Risk; International Coordination Challenges; Energy Transition Uncertainty.

³⁶ This application of vignette analysis to policy analysis and political decision making builds on the approach outlined in Bakker et al. (2014).



Precision relief

Recognising unequal impacts of disruptions to energy supply across provinces and sectors, such as coastal manufacturing hubs or agribusiness, Beijing authorises selective and targeted drawdowns to alleviate acute shortages for factories, transportation and key regional markets. This approach balances national energy security with the need to prevent economic bottlenecks and allows China to maintain growth and stability in its most competitive sectors while keeping broader reserve depletion in check. The government's strategic focus on targeted relief is lauded by business leaders and regional officials, who see it as a lifeline, but it draws criticism from those outside the prioritised zones. International partners interpret the move as China flexing its reserve-management capabilities to reinforce domestic advantage, while also hinting at its capacity for finely tuned economic interventions in a crisis.

Expert assessment

The experts found the precision relief scenario highly realistic, noting that China's policy system enables targeted interventions by directing SPR resources to specific regions or sectors in crisis. They highlighted that China already differentiates policy across provinces and industries, has the administrative capacity for such selective actions, and would be motivated to prevent major economic disruptions in critical areas while minimising overall reserve depletion. This capacity to direct SPR deployment may give China an advantage over other countries that take a more market-based approach to SPR releases.



Strategic energy diplomacy

Amid acute fuel shortages and market instability across Asia and the developing world, China leverages its SPR to expand its regional and global influence. Offering vital oil and refined products – particularly jet fuel – to selected countries facing crises, Beijing provides energy relief at below-market rates, but ties these lifelines to expectations of diplomatic alignment or economic cooperation. To deepen these relationships and promote greater monetary sovereignty, China pioneers SPR-backed deals denominated in yuan, testing the potential for currency diversification and providing an attractive alternative to countries vulnerable to dollar-based sanctions or financial pressure. This strategy not only secures new partners and strengthens China's position as a leading energy supplier, but also challenges Western influence, especially in the Global South.

Expert assessment

The experts believed it is possible that China would use its SPR to provide jet fuel and other refined products as diplomatic incentives, consistent with its preference for offering 'carrots' rather than imposing punishments in international engagement. While such conditional energy aid is likely to help China secure strategic partnerships in Asia and the Global South, recipient countries may remain cautious about becoming too dependent.

China's approach to its SPR is likely to remain cautious and conservative over the short to medium term, with the country only rarely and carefully releasing SPR volumes, even during episodes of elevated market volatility or external shocks. This is consistent with the current policy focus on domestic stability and resilience. This cautious stance is reinforced by the continuing opacity surrounding China's SPR data – which contrasts with the transparency requirements of IEA members – and the central government's desire to keep its options open.

Over the medium to long term, China's energy landscape will evolve as it continues to electrify its economy and scale up its nuclear and renewable energy generation capacity. A reduced economic petroleum intensity would mean that China needs fewer petroleum reserves to cover its needs over a prolonged period of time. At the same time, experts agreed that it is unlikely China would deliberately reduce the size of its SPR in the medium term. Even as the civilian energy system transitions to low-carbon sources, military applications will likely remain dependent on liquid fuels for longer. Preparedness for potential conflict underpins the need for substantial strategic oil stocks regardless of gains in renewables or electrification.

The experts agreed that the combination of these two trends could give China the policy space to pursue a more ambitious use of the SPR as a geoeconomic instrument. For example, China could use targeted SPR releases to support strategic partners, as leverage in bilateral relationships or to build influence, especially in the Global South or within the Association of Southeast Asian Nations (ASEAN).

The other vignettes on the longlist were also debated. For instance, using aggressive SPR releases to undercut global competitors and gain industrial advantage, for example by subsidising petrochemicals or manufacturing exports, was deemed improbable. Experts pointed to the risks of domestic backlash, the existing overcapacity in Chinese industry, and concerns about provoking defensive reactions in key markets such as the EU and US. Deploying the SPR as part of coordinated international releases was recognised to be possible in theory, particularly in the context of a severe global crisis, but China's historical preference for unilateralism means such moves remain unlikely except in truly extreme circumstances.

The experts also examined the appeal for China of using the SPR in a broader geoeconomic strategy, such as advancing renminbi internationalisation. SPR-backed oil deals denominated in yuan are already occurring, particularly with smaller or sanctioned trading partners, but the experts noted that meaningful progress towards a true 'petroyuan' system would require much deeper domestic reforms, including significant capital account and financial liberalisation. At present, such deals provide useful signalling and incremental progress, but are not expected to catalyse a dramatic shift in global energy trade currency dynamics without broader systemic changes.³⁷

Finally, the concept of leveraging the SPR to build regional or global alliances, for example through ASEAN energy diplomacy or a China-led alternative to the US-oriented IEA coordination mechanism, was considered intriguing but challenging. The experts noted that while China has an incentive to be seen as a 'team player' and to increase its regional leadership, realising such a coalition would require sustained diplomatic effort and greater transparency. ASEAN member states may remain wary of Chinese intentions and reluctant to tie their own energy security to Chinese decision making without robust institutional mechanisms and trust.

In summary, the experts underscored that the evolution of China's SPR policy over the coming years is likely to proceed incrementally, but with the potential for more assertive geoeconomic uses as China's energy landscape and confidence grows. The SPR is poised to remain a key instrument of national resilience, even as its potential role as a tool of external influence expands.

³⁷ See also Gang (2026).

6. Conclusion

The RAND subject-matter experts who took part in the workshop agreed that China's SPR has grown into a substantial and increasingly sophisticated system, with a capacity rivalling the world's largest oil importers and backed by a recently strengthened legal and institutional framework. The baseline scenario – strategic caution – was identified as the most likely, with China prioritising energy security, domestic resilience and stability through conservative, sparing SPR releases. This approach reflects Xi Jinping's leadership priorities and China's traditional risk aversion in managing critical resources.

At the same time, experts viewed the precision relief scenario, where SPR resources are selectively released to specific provinces or sectors facing acute shortages, as highly realistic. China's established tradition of differentiated policy and administrative flexibility supports such targeted interventions, enabling authorities to address bottlenecks in strategic areas while preserving broader reserve integrity. In contrast, the US approach is more centralised and transparent, prioritising broad, market-based SPR releases to achieve national-scale impact rather than precision targeting within its own economy. The experts also noted that conditional energy diplomacy, such as providing jet fuel or refined petroleum products to secure influence and partnerships, remains a plausible instrument, especially in Asia and the Global South.

Looking ahead, the panel noted that China's ongoing energy transition – with increased electrification, renewables and nuclear power – may diminish its dependence on oil over the course of the next two decades. This shift could allow Beijing to use the SPR more assertively for diplomatic or geoeconomic gains, especially in crises or when cultivating strategic relationships. Overall, the workshop underscored the dynamic role of China's SPR, suggesting its management will be increasingly consequential for global energy security and international power balances, and highlighted the importance of ongoing monitoring and analysis.

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Annex A. Annex

For the development of the scenarios, the project team referred to six enabling drivers, which could encourage China to use the SPR in a geoeconomic way, and six restricting drivers, which could limit China's use of the SPR as a geoeconomic tool. These drivers are outlined in the tables 4 and 5 below.

Table 4. Enabling drivers

Driver name	Driver is strong	Driver is medium	Driver is weak
Sanctions and Signalling Resilience	China faces ongoing, severe Western sanctions or credible threats of supply disruption and must visibly demonstrate resilience.	Severe disruptions are possible or moderate but not likely; some resilience signalling is needed but not prioritised.	International environment is cooperative or stable; little threat of disruptions or embargoes.
Global Energy Market Influence	China has the capacity, intent and opportunity to move or set global oil prices with clear market interventions.	China participates in market signalling but is one of several influential players; interventions affect regional or short-term prices.	China's market interventions have minimal global impact; little incentive or capacity for price setting.
Energy Portfolio Diversification	China has robust alternatives (renewables/nuclear), making it less reliant on oil and allowing bold SPR use.	Alternatives are growing but not yet dominant, allowing some flexibility but not total independence from oil.	China remains highly dependent on oil; little leeway for bold SPR action.
Regional Security Signalling	Regional tensions (e.g. Taiwan, South China Sea) are high and energy measures are clearly linked to security signalling or deterrence.	There are ongoing, but not acute, regional disputes that require occasional energy signalling.	Regional environment is calm; energy policy is not linked to security messaging.
Strategic Industry Prioritisation	Critical sectors (e.g. manufacturing, agriculture) face acute stress and require government support through energy flows.	Some industries are stressed, but require only targeted or partial intervention.	Industries are healthy; no special need for supportive SPR releases.
Strategic Partnerships with the Global South	China seeks to actively build influence and goodwill in the Global South with frequent supply deals and SPR aid.	Engagement with developing countries is ongoing but not aggressive; selective aid or swaps occur.	Minimal outreach; SPR is used primarily for domestic priorities.

Table 5. Restricting drivers

Driver name	Driver is strong	Driver is medium	Driver is weak
Long-term Security of Supply Challenges	Chronic concerns about energy import dependence and supply continuity; high motivation to conserve reserves.	Some anxiety about future scarcity, but not overriding; balanced consideration in reserve policy.	Supply security is not a major worry; imports are stable and plentiful.
Domestic Stability and Public Discontent	High risk or presence of public unrest/inflation if reserves are prioritised externally; domestic priorities dominate.	Some public dissatisfaction, but not at crisis level; balanced with other interests.	Public accepts government energy policy; little visible discontent.
Weakening Economic Conditions	Growth is sluggish, inflation and unemployment are high, and consumer sentiment is poor; economic crisis dominates decision-making.	Economic signals are mixed; policymakers are cautious but not alarmed.	Economy is robust and growing; little economic constraint on reserve use.
Market Volatility Risk	Oil prices are extremely volatile; market swings make large SPR actions risky and potentially ineffective.	Moderate price changes and some uncertainty; manageable but notable risk.	Oil prices are stable and predictable; timing and scale of SPR use is straightforward.
International Coordination Challenges	Major challenges in aligning SPR policy with other countries; cooperation is slow or politically fraught.	Some misalignment or technical issues but coordination is possible with effort.	Coordination is easy and effective; China can act seamlessly with partners.
Energy Transition Uncertainty	Uncertainty about pace and scale of shift to renewables/nuclear casts doubt on reserve planning and usage.	Transition is happening but somewhat predictable; some risk to reserve management.	Energy transition is slow or well-anticipated; little uncertainty affects SPR policy.